

Carlos De Manuel Vicente

Data & Analytics Engineer

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I build the data layer a business runs on — at a global automotive manufacturer, from top-management reporting down to shop-floor apps, across 20+ plants in North America, APAC and EMEA — on Snowflake, dbt, Python, Prefect, AWS and Power BI. I start from the process and the decision behind a number, simplify before automating, and ship small, tested increments.

EXPERIENCE

Data & Analytics Engineer (official title: Digital Transformation Specialist) Jan 2025 – Present
RESRG Automotive · global automotive manufacturer Liria, Spain (hybrid)

- Replaced manual Excel trackers across **20+ plants** (NA, APAC, EMEA) with self-serve data products in Streamlit and Power BI — from the executive summary **200+ people** open daily to inventory apps on the shop floor, across purchasing, finance, supply chain and production.
- Built near-real-time analytical models for a warehouse management system over **800M+ rows** in Snowflake, refreshed by event-driven Prefect flows that fire when upstream Fivetran HVR replication lands, with the data engineering team.
- Set one definition for **130+ operational KPIs** through a KPI-governance app built on dbt, with tests on every model and pipeline observability in Prometheus.
- Re-engineered the end-to-end refresh with CI/CD, cutting runtime **~98%** and compute cost **~90%**.
- Forecast plant electricity consumption from production variables for the purchasing team, joining utility invoices with production and financial data so usage normalises against sales.

Data Scientist (internship) Nov 2023 – Jul 2024
Biomedical Data Science Lab, ITACA — UPV Valencia, Spain

- Built an end-to-end Python framework and preprocessing pipelines to evaluate healthcare ML models for fairness, robustness and explainability (XAI).

PROJECTS

Job Market Intelligence · carlosdmv7.github.io/personal-portfolio/projects/job-market-intelligence
Daily pipeline reading EU data-job postings from six free APIs; LLM enrichment batched to read **23 → 180** postings a day on the same free quota; dbt medallion on MotherDuck (9 models, 53 tests); Streamlit app ranking roles against a CV. Prefect Cloud deployment, €0 a month.

Spanish Housing Radar · carlosdmv7.github.io/personal-portfolio/projects/spanish-housing-radar
Weekly scrape of València listings, each scored against an empirical-Bayes blend of its barrio and district benchmark; dbt medallion (19 models, 162 tests), Prefect Cloud, Docker, Streamlit.

SKILLS

Warehouse & modelling	Snowflake, dbt, DuckDB, Kimball dimensional modelling
Languages	Python, SQL, Jinja, DAX
Platform & orchestration	AWS, Prefect, Airflow, GitHub Actions (CI/CD), Git, Azure DevOps, Docker, Make
Quality & observability	dbt tests, pytest, Pydantic, ruff, mypy, Prometheus
BI & data apps	Power BI, Streamlit, Microsoft Fabric
AI & ML	Gemini and Claude APIs, LLM evals, scikit-learn, SHAP

EDUCATION

MSc, Data Analysis, Process Improvement & Decision Support Engineering — Universitat Politècnica de València (UPV) 2024 –2025

BSc, Biomedical Engineering (ICT specialisation) — Universitat Politècnica de València (UPV) 2020 –2024
Thesis graded 10/10 · distinction for academic performance (top 25%)

CERTIFICATIONS & LANGUAGES

AWS Certified Data Engineer – Associate and dbt Certification (both in progress, 2026)
Spanish (native) · English (C1, Cambridge, 194)